

11.15 Do Now: Introduction to Periodic Functions — Solutions

1. For $f(x) = 3 \sin(2x) - 1$, comparing to $f(x) = a \sin(b(x + c)) + d$: $a = 3$, $b = 2$, $c = 0$, $d = -1$.

(a) amplitude $= |a| = 3$.

(b) period $T = \frac{2\pi}{b} = \frac{2\pi}{2} = \pi$.

(c) midline $y = d = -1$.

(d) maximum value $= d + |a| = -1 + 3 = 2$.

2. For $g(x) = 4 \cos\left(\frac{\pi}{6}(x - 2)\right) + 5$, comparing to $g(x) = a \cos(b(x + c)) + d$ (with $c = -2$):
 $a = 4$, $b = \frac{\pi}{6}$, $c = -2$, $d = 5$.

(a) amplitude $= 4$.

(b) period $T = \frac{2\pi}{\pi/6} = 12$.

(c) midline $y = 5$.

- (d) g is a maximum when the argument of $\cos$ is 0 (or any multiple of 2π). Setting

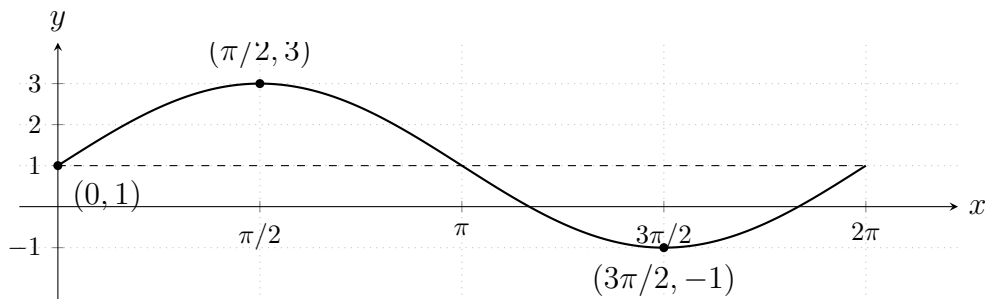
$$\frac{\pi}{6}(x - 2) = 0 \implies x = 2.$$

Since $x = 2 \geq 0$, the smallest such x is $\boxed{x = 2}$.

3. $y = 2 \sin x + 1$. Amplitude 2, period 2π , midline $y = 1$.

Key features for the sketch:

- y -intercept: $f(0) = 2 \sin 0 + 1 = 1$.
- Maximum: $y = 1 + 2 = 3$ at $x = \frac{\pi}{2}$.
- Minimum: $y = 1 - 2 = -1$ at $x = \frac{3\pi}{2}$.
- Zero crossings (where $\sin x = -\frac{1}{2}$): $x = \frac{7\pi}{6}$ and $x = \frac{11\pi}{6}$.



4. Given $f(x) = a \sin(b(x + c)) + d$ with $\max = 7$, $\min = 1$, period $T = 8$, and first maximum at $x = 1$.

Step 1 (amplitude and midline).

$$a = \frac{\max - \min}{2} = \frac{7 - 1}{2} = 3, \quad d = \frac{\max + \min}{2} = \frac{7 + 1}{2} = 4.$$

Step 2 (period factor).

$$b = \frac{2\pi}{T} = \frac{2\pi}{8} = \frac{\pi}{4}.$$

Step 3 (phase from the first maximum). At a maximum, $\sin(b(x + c)) = 1$, so $b(x + c) = \frac{\pi}{2}$. Substituting $x = 1$:

$$\frac{\pi}{4}(1 + c) = \frac{\pi}{2} \implies 1 + c = 2 \implies c = 1.$$

Answer. $a = 3$, $b = \frac{\pi}{4}$, $c = 1$, $d = 4$. That is,

$$f(x) = 3 \sin\left(\frac{\pi}{4}(x + 1)\right) + 4.$$

Check. $f(1) = 3 \sin(\pi/2) + 4 = 7$ ✓; $f(5) = 3 \sin(3\pi/2) + 4 = 1$ ✓ (minimum half a period later).